# Make XLS from CSV

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# Summary

- Currently, only XLSX generation is supported and that's fine. Adding the macro insertion to generate XLSM is trivial and not part of this ask
- Currently, only this use model is supported :

```
$ > python3 make_xlsm.py test_mk_xlsm.csv
```

# this generates the **Corners** sheet with all data, and the **EC Table** and **EC w Links** sheets which contain min,max data. The difference between the latter two is that the **EC w Links** has cells that are links

- We want to add this use model :

```
$ > python3 make_xlsm.py input.csv [filter-or-breakout-spec1] [FOBS2] ..
```

# Example File : test_mk_xlsm.csv

- L, VDD, Corner, Point, temperature are “variables”
- A, B, C, D are “outputs”
- These are separated by the “Pass/Fail” column

- It is easier to see this looking at the test_mk_xlsm_1.xlsx file

- From the script’s point of view, we make no distinction (between outputs and variables) because the function is always to filter out rows based on criteria

# Filter or Breakout Specifier

- Breakout means : use the specification to generate a separate sheet for each value of that variable that meets the specification
- Filter means choose or eliminate rows from the starting dataset based on the specified criterion
- Can be combined - E.g. “use the L values to generate a separate sheet for each one, with data for temperature=27 (only)” (that is, if you look at the example data, temperature data exists for values 0,27,100 and the user is asking for additional sheets (in this case you will have six additional sheets) which each having data for a unique value of L and only the rows for which temperature is 27)

# Command Line Syntax

- If necessary, the specifiers will be enclosed in quotes to hide them from the shell (which would otherwise process characters like ! and > and <)
- Eg : `$ python3 make_xlsm.py in.csv ‘L,>4.5;VDD<1’`

# Specifier Examples

| # | Specifier | Meaning |
|---|---|---|
| 1 | L | For each unique value of L, create a separate EC Table and EC w Links type sheet. Since the example data has values 4,5,6.25, the sheets will be named “EC L 4”, “ECL L 4” |
| 2 | L,condition | The **comma after the variable name** indicates that this is a breakout spec - you must use the condition to select values and then generate additional sheets for each value |
| 3 | L;VDD | This is illegal - only one variable is allowed to specify breakout - other variables appearing in the specifier are for filtering |
| 4 | L;VDD>1.1 | Breakout the L values and only use data for which VDD > 1.1 |
| 5 | L,<6;VDD<0.99 | Breakout the L values that are less than 6 (that is, for the example dataset, this would result in four additional tables, two each for L=4 and L=5. Further, filter out data that does not meet VDD < 0.99 |

# Specifier Examples

| # | Specifier | Meaning |
|---|---|---|
| 6 | L=5;VDD>0.9 | This is purely filtering. In filtering the conditions are always combined in AND fashion - there is no support for OR Here we are choosing data (and creating the two additional tables) for which L= 5 AND VDD > 0.9 |
| 7 | L!5;VDD>0.9 | Also pure filtering, here we are choosing data for which L is not equal to 5 and VDD is > 0.9 |
| 8 | L!5,6.25 | Pure filtering - the values to exclude are specified |
| 9 | L,!5 | Breakout - value to exclude specified (so in this case, given the data - 4,5,6.25 are the values, it’s the same as L,=4,6.25 |
| | | |

# Positioning the "Additional" Sheets

- "Additional" sheets are what you get compared to the default case that has only **Corners, EC Table** and **EC w Links** sheets
- The additional "plain" **EC Table** style sheets must be added between the default **EC Table** sheet
- The additional **EC w Links** style sheets must be added after the default **EC w Links** sheet

# Naming the Sheets

- Use the filter specifier as is. If necessary, replace ! with ~
- for breakout, **EC Table** → EC, then append variable name and value and semicolon and condition. **EC w Links** → ECL, then (same as for EC Table) (E.g. **EC L 6.25** and **ECL L 6.25** for the breakout case)
- For filter, specify the filter as is, replacing ! with ~ if necessary (Eg **EC L>5;VDD5;VDD<0.99** and **ECL L>5;VDD<0.99**)

# More Command Line Examples

- `$ > python3 make_xlsm.py test_mk_xlsm.csv L!5 L!5,625`

in this case, there will be four additional sheets compared to the default case : "EC L~5", "ECL L~5" and "ECL L~5,6.25" and "ECL L~5,6.25“

- `$ > python3 make_xlsm.py test_mk_xlsm.csv L=5 VDD=0.778`

in this case, there will be four additional sheets compared to the default case : "EC L 5", "ECL L 5" and "EC VDD 0.778" and "ECL VDD 0.778"

# Inventory

| # | Name | Description |
|---|---|---|
| 1 | make_xlsm.py | Starter python script needing enhancement |
| 2 | test_mk_xlsm.csv | Input CSV data |
| 3 | test_mk_xlsm_1.xlsx | Copy of the test_mk_xlsm.xlsx that is produced from<br>`$ > python3 make_xlsm.py test_mk_xlsm.csv` |
| 4 | test_mk_xlsm_L_6p25.xlsx | Copy of the test_mk_xlsm.xlsx that is DESIRED from<br>`$ > python3 make_xlsm.py test_mk_xlsm.csv L=6.25` |